The Dispersion-Witness Frontier for One-Scale Yang–Mills Control

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Abstract

We compose ten variance-aware phases. The refined split closes in the exact dependent Gaussian block, supplies adaptive and normalized-repair interfaces, and carries periodic center flux. The remaining physical obstruction is an explicit nonlinear repair/comparison inequality uniform in retained state and volume.

1 Strongest combined theorem

Theorem 1. *On the open 2^4 projected-Gaussian block at $t = 3{\cdot}10^{-6}$, there is a nonempty interval of RMS staple-dispersion thresholds for which both hold: (i) the low-dispersion finite-time conditional channel meets its declared ceiling; (ii) the high-dispersion determinant-Chernoff polymer has rooted free energy exceeding the required target. No independence is used.*

Proof. Stage 32 proves (i) up to RMS radius 0.00717145149. Stage 35 proves (ii) from RMS radius 0.00386371990. The strict inequality between endpoints gives the interval. □

The run also proves: covering failure costs η^2; canonical curvature witnesses have exact congestion four; action-only plaquette counting is invalid after radial entropy; shared-uniform adaptive coupling is exact; weighted repair congestion is exact; five-channel budgets must sum; and the periodic center carrier has six independent flux labels.

2 First open physical inequality

For every finite volume Λ, positive-mass retained history h, and owned channel c, construct a gauge-covariant nonlinear comparison or repair T_c such that

$$\sum_c \left(e^{\ell_c} q_{\mathrm{low},c}(\Lambda, h) + C_{T_c}(\Lambda, h) \right) \le 0.002430277501841998,$$

uniformly, with multiplicity, Jacobian, state weight, chart transitions, center flux, and topology retained. This would instantiate the abstract adaptive premise and normalized partition ratio.

Remark 1. *That inequality is open. The Gaussian closure is a rigorous restricted-model theorem, not physical four-dimensional basin entry. Integer bundle topology, RG iteration, continuum construction, Osterwalder–Schrader reconstruction, and the Hamiltonian mass gap remain open. The strongest level is C2.*